F16AG424-VS
VARIABLE SPEED DRIVE

SYSTEM INTERCONNECTION DIAGRAM 2000-EL005-B02-10001

11,000V
RD WH BU
FCF 100A
QAC 400A
BCT1 50 / 1A 2.5VA 5P20
BCT2 50 / 1A 2.5VA 5P20
BCT3 50 / 1A 2.5VA 5P20
BCN 50 / 1A 1.5VA 10P10
BAD
PFV1 RD RD RD
QCE 400A
EARTH BAR
X4-1 X4-2 X4-3 X4-4 X4-5 X4-6 X4-7 X4-8 X4-11 X4-12 X4-13
2000-EL003-F16-10016 SHEET 14

L1 L2 L3
LINE SIDE INCOMING SUPPLY SHEET 6

T1 T2 T3
OUTGOING MOTOR SUPPLY SHEET 6

CABLE SCREEN EARTHED
M 3~
F16-AG-424
ROUGHER FLOTATION CELL 1 AGITATOR
3,300V
500 kW

F16-AG-424
MOTOR HEATER TERMINAL BOX
F16-HT-544
0.3kW
X2 54 53
F16HT544-L
F16HT544-N
ANTI-CONDENSATION HEATER

MOTOR SPACE HEATER CIRCUIT SHEET 6
X81 (PART OF)
RTD TEMPERATURE MONITOR RELAY SHEET 6

F16-AG-424
MOTOR RTD TERMINAL BOX
X81
PT100 RTD INPUT CH1: + 13, COM 14, - 15
PT100 RTD INPUT CH2: + 16, COM 17, - 18
PT100 RTD INPUT CH3: + 19, COM 20, - 21
PT100 RTD INPUT CH4: + 22, COM 23, - 24
PT100 RTD INPUT CH5: + 25, COM 26, - 27
PT100 RTD INPUT CH6: + 28, COM 29, - 30
PT100 RTD INPUT CH7: + 31, COM 32, - 33
PT100 RTD INPUT CH8: + 34, COM 35, - 36
F16AG424A-RD
F16AG424A-WH
F16AG424A-BK
F16AG424B-RD
F16AG424B-WH
F16AG424B-BK
F16AG424C-RD
F16AG424C-WH
F16AG424C-BK
F16AG424D-RD
F16AG424D-WH
F16AG424D-BK
F16AG424E-RD
F16AG424E-WH
F16AG424E-BK
RTD
MOTOR WINDING RTD (RD PHASE) (TD1)
MOTOR WINDING RTD (WH PHASE) (TD3)
MOTOR WINDING RTD (BU PHASE) (TD5)
MOTOR BEARING RTD (DRIVE END) (BD10)
MOTOR BEARING RTD (NON DRIVE END) (BD20)

F16-SW-011 11kV FEEDER START/STOP CONTROL 2000-EL003-F16-10016 SHEET 7 3A & 3B
X3 3 1139 X2 67
X3 17 2703 X2 68 K41 K41
MV CONTACTOR ENABLE CIRCUIT SHEET 6

F16-SW-011 11kV FEEDER VSD HEALTHY 2000-EL003-F16-10016 SHEET 10 4A
X2 4 1179 X2 63 K41 K41 K41
X2 25 1199 X2 64
MV INPUT ENABLE CIRCUIT SHEET 6

F16-SW-011 11kV FEEDER VSD TRIP 2000-EL003-F16-10016 SHEET 10 3A
X2 3 1219 X2 65
X2 24 1239 X2 66 K41 K41 K41
MV INPUT TRIP CIRCUIT SHEET 6

F16-SW-011 11kV FEEDER ARC FLASH TRIP 2000-EL003-F16-10016 SHEET 8 5A & 5D
X2 56 1259 X2 70
X2 44 1279 X2 69
A30 (PART OF) X2 15 TRIP X2 16
ARC FLASH DETECTION TRIP SIGNAL SHEET 6

F16-DB-101
2000-F16-TLD-EL-10054
RD WH BU BK
R11 W11 B11 25A
415VAC
EARTH BAR
F16DB101-R11
F16DB101-W11
F16DB101-B11
F16DB101-BK11
X4 1 2 3 4 PE1
VSD AUXILIARY POWER SUPPLY SHEET 6

F16-DB-201
2000-F16-TLD-EL-10057
RD BK
R4 10A
240VAC
EARTH BAR
F16DB201-R4
F16DB201-BK4
X4 5 6 PE2
VSD UPS POWER SUPPLY SHEET 6

TO FLOTATION 11kV SWITCHROOM
F16-COM-001 COMMUNICATION RACK
MODBUS ETHERNET SWITCH
F16-ETH-001
2000-F16-TER-IC-10002
CAT6 — CAT6 — CAT6 — CAT6 — CAT6 — CAT6 — CAT6 — CAT6
X47 ETHERNET ADAPTOR RJ45
A2 (PART OF) DI-2B 9
X2 13 14

SAFE TORQUE OFF CIRCUIT SHEET 6
X86 1 2 3 4
F16AG424VS-1409
F16AG424VS-1419
F16AG424VS-1429
F16AG424VS-1439

F16AG424-LCS
LOCAL CONTROL STATION
X1 1 2 3 4
LOS
X1 5 6 E
START
F16AG424VS-1459
F16AG424VS-1469
24VDC CONTROL SHEET 6

NOTES

1. WIRE NUMBER FERRULES ON TERMINAL STRIP SHALL BE PREFIXED WITH EQUIPMENT NUMBER e.g. F16AG424-1200, UNLESS OTHERWISE NOTED.
2. EACH WIRE SHALL BE IDENTIFIED AT BOTH ENDS WITH FERRULES.

LEGEND

- 11kV SWITCHBOARD TERMINAL
- VSD TERMINAL
- FIELD TERMINAL
- DCS OR I/O TERMINAL
- MARSHALLING TERMINAL
- DEVICE TERMINAL

PANEL A05

2000-F16-SCM-EL-10109_Rev1

F16-SW-011 F16-AG-424 scavenger flot. cell 1 agitator
VSD fdr - schematic diagram

O:\CAD PROJECTS\235 NORTHERN STAR RESOURCES LTD\23517 KCGM GROWTH PROJECT\03 DELIVERABLES\ELVF16 - ROUGHER & SCAVENGER FLOTATION\2000-F16-SCM-EL-10109.DWG 09SEP25 6:39AM HOTSEAT04